

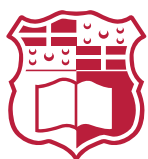
Evaluating the Quality of the LAION-2B Dataset: Insights into Large-Scale Image Dataset Assessment

Jerome Agius

Supervisor: Dr Dylan Seychell

May 2026

*Submitted in partial fulfilment of the requirements
for the degree of Master of Science in Artificial Intelligence [Taught and
Research (Mainly by Research)] (M.Sc.(Melit.)).*



L-Università ta' Malta

**Faculty of Information &
Communication Technology**

Abstract

Generative Artificial Intelligence (AI) systems have enabled the large-scale creation of images, with these technologies becoming increasingly accessible to the public. However, the development of these systems is largely dependent on the existence of massive web-scraped datasets serving as training data, such as Re-LAION-5B, which forms the focus of this research alongside the Stable Diffusion v1.5 model trained on it. Due to the nature of web-scraped training data, there has been increased concern regarding the ethical implications imposed, particularly related to bias and unintended correlations embedded within the training corpus. Specifically, non-semantic or spurious features present within the training dataset may serve as shortcuts for various downstream tasks, negatively impacting model behaviour and fairness outcomes. While these shortcuts can emerge for any task, demographic attributes represent a particularly concerning application, alongside other attributes unrelated to the inherent semantic content of the dataset.

This thesis investigates the demographic bias within the Re-LAION-5B dataset and resultant generative images produced by the Stable Diffusion v1.5 model, with a direct focus on identifying spurious features within these two components and determining whether the generative model amplifies, alters, or mitigates those spurious features identified within the respective training corpus. The study adopts a classifier-based approach derived from prior research, in which discriminative models are trained to distinguish between Re-LAION-5B and Stable Diffusion v1.5 generated images. These transformations systematically reduce the semantic information available to the model and force reliance on specific low-level image features, thereby exposing the presence of dataset specific spurious patterns that can be exploited by the model for image source identification.

Building upon this classifier based analysis, the relevant features used in the discrimination task are further annotated in terms of gender and skin-tone attributes. This secondary analysis investigates whether the previously identified dataset specific spurious features demonstrate correlations with these sensitive demographic attributes, allowing for the detection of harmful features that may serve as model shortcuts for deriving gender or skin-tone classifications. To further contextualise these findings and assess the real world implications of observed biases, the demographic distributions of retrieved and generated images across 200 occupation categories are compared against real world demographic statistics, revealing patterns of representation bias and potential amplification effects in the generative model.

Finally, based on the spurious features identified in the prior sections, this thesis implements and evaluates a combined set of inference-time interventions for debiasing

the Stable Diffusion v1.5 model without requiring full model retraining. These interventions demonstrably reduce spurious demographic correlations across low-level, geometric, and semantic representations, confirming that inference-time debiasing can produce measurable and substantive departures from the demographic and spurious feature profile of standard generation. By explicitly linking dataset characteristics, spurious feature learning mechanisms, and generative bias outcomes, this work contributes both methodological insights for evaluating large-scale multimodal datasets and practical interventions for the development of fairer text-to-image generation systems.

Acknowledgements

I wish to express my sincere appreciation to my supervisor Dr. Dylan Seychell, for his guidance and support throughout the completion of this Master's thesis. His expertise and mentorship played a crucial role in navigating the challenges encountered during this endeavour. Furthermore, I am deeply grateful to my parents, Reno and Graziella, and my brother Julian, for their encouragement and support throughout this academic pursuit.

Contents

Abstract	i
Acknowledgements	iii
Contents	iv
1 Introduction	1
2 Background & Literature Review	2
3 Methodology	3
3.1 Image Retrieval and Dataset Construction	3
3.2 Demographic Annotation	3
3.3 Image Transformation Pipeline	4
3.4 Inference-Time Debiasing Pipeline	4
4 Evaluation	5
4.1 Demographic Distribution Analysis	5
4.2 Dataset Classification	5
4.3 Demographic Artifact Probing	6
4.4 Debiasing Evaluation	6
5 Conclusions and Future Work	8

1 Introduction

Text-to-image diffusion models, Stable Diffusion [1] in particular, have achieved widespread deployment while relying on massive web-scraped corpora such as Re-LAION-5B [2], whose automated construction introduces systematic biases that reflect societal inequities [3]. Existing research documents demographic bias in generative outputs [4, 5] but treats models as black boxes, ignoring the *spurious features* — low-level non-semantic characteristics such as colour statistics, texture, and edge distributions — that correlate with demographic attributes and enable biased shortcuts [6, 7]. This thesis addresses these gaps through a mechanistic investigation of spurious features in Re-LAION-5B and SDv1.5, applying established methodologies to expose low-level features that distinguish dataset sources and correlate with demographic attributes, then evaluating targeted inference-time interventions without model retraining. Five objectives structure the work: (O1) dataset acquisition and annotation across 200 ISCO-08-aligned occupations; (O2) spurious feature detection via classifier-based transformation analysis; (O3) demographic correlation assessment against real-world statistics; (O4) inference-time debiasing of SDv1.5; and (O5) synthesis of how training data biases propagate through the generative process.

2 Background & Literature Review

Re-LAION-5B comprises approximately 5.85 billion image-text pairs harvested via Common Crawl [2], re-released with improved safety filtering [9]; its LAION-Aesthetics v2 5+ subset constitutes the training corpus for SDv1.5 [10]. SDv1.5 is a Latent Diffusion Model [1] that denoises in a compressed latent space via a DDPM U-Net conditioned on CLIP text embeddings [11, 12]; higher classifier-free guidance scales amplify demographic stereotypes [13]. Its fully documented training lineage makes it uniquely suited for mechanistic bias auditing, unlike later models such as SDXL [14] that rely on proprietary corpora. Zeng et al. [7] operationalise spurious feature analysis through a transformation-based classification framework in which each transformation, spanning pixel shuffle, patch shuffle, mean RGB, frequency filtering, Canny edge, SAM contour, monocular depth, semantic segmentation, object detection, and image captioning, isolates a distinct visual property while discarding all others, revealing that dataset identity is redundantly encoded across colour, texture, geometry, and semantic content simultaneously and that generative models reproduce training data biases in their outputs. Meister et al. [6] extend this to demographic attributes via a systematic occlusion-based probing protocol, establishing that gender artifacts are distributed across the entire image structure rather than localised to the subject, undermining fairness-through-blindness assumptions. This thesis extends Zeng et al.’s framework to a three-class Re-LAION-5B / SDv1.5 / COCO setting and applies Meister et al.’s occlusion protocol to both retrieved and generated corpora.

Occupation prompts drawn from the ISCO-08 taxonomy [15] provide an interpretable probe for demographic priors [4, 5, 16]. Demographic annotation uses automated classifiers applied to segmented facial skin regions to prevent background shortcut learning [17], with skin tone operationalised via the ten-point Monk Skin Tone scale [18] and evaluated against CCv2 [19] and FACET [20]; automated annotation introduces structured measurement error [21]. Inference-time debiasing is applied via ITI-GEN [22], which learns inclusive text-space tokens grounded in demographic reference images, composed with ControlNet OpenPose [23] for structural conditioning and Chamfer-based colour alignment [24] to neutralise colour-level spurious priors.

3 Methodology

3.1 Image Retrieval and Dataset Construction

Re-LAION-2B-en-research-safe was processed across four uniformly sampled shards, yielding 65,552,809 URL-caption pairs of which 43,491,938 images were successfully retrieved (link decay rate 33.7%). CLIP ViT-L/14 embeddings were computed, L2-normalised, and indexed per shard using FAISS IndexFlatIP for exact cosine similarity retrieval. A set of 200 ISCO-08-aligned occupations was retrieved using the template "A photo of a/an {profession}" with gender-neutral phrasing, filtered via a negative prompt dominance criterion, and passed through a multi-stage structural validation pipeline requiring exactly one detected face and person with verified spatial association and successful MediaPipe FaceMesh landmark extraction. YOLOv12l-Face was selected as face detector (AP@0.5 0.905, highest small-face recall of four candidates). The top 1,000 structurally valid images per occupation were retained, yielding 200,000 images at a mean acceptance rate of 33.51%. A corresponding 200,000-image generation corpus was produced using SDv1.5 with LCM-LoRA acceleration, selected for throughput (1.5 s/image versus 5.6 s for standard SDv1.5) from three LAION-traceable configurations. A full-body profession-conditioned prompt with neutral professional framing was used throughout; classifier-free guidance was fixed at $s=1.0$ to establish a conservative lower bound on bias free from inference-time amplification. Generated images passed through the same structural validation pipeline, achieving a mean acceptance rate of 66.38%.

3.2 Demographic Annotation

All images underwent FaceMesh face segmentation producing background-removed crops to prevent background shortcut learning. Gender was annotated using the Realistic Gender Classifier, selected from five candidates for its highest CCv2 face-only accuracy (88.6%) and smallest male-female gap (6.2 pp versus up to 40.2 pp for alternatives). Skin tone was estimated using a VGG-16 MST10 regression model selected via factorial ablation across output formulation, colour space, label granularity, and background treatment, with predictions collapsed into Light (MST 1–3), Mid (MST 4–7), and Dark (MST 8–10) bins. All labels are treated as perceived-attribute proxies. Distributions were contextualised against BLS and ILOSTAT reference statistics; a race-to-skin-tone mapping via FairFace was abandoned due to mid-bin collapse.

3.3 Image Transformation Pipeline

The pipeline serves two objectives: dataset classification (ConvNeXt-Tiny distinguishing Re-LAION-5B, SDv1.5, and COCO per transformation) and demographic artifact probing (predicting gender and skin tone from isolated visual channels). Adding COCO as a third class enables per-class accuracy attribution, localising which dataset carries each spurious signal. Demographic probes use deliberately simple models: logistic regression for low-dimensional inputs and ResNet-50 for spatial inputs, following Meister et al.

Transformations span three levels. *Structural*: monocular depth (Depth-Anything-V2 ViT-S), Canny edge detection, low-pass and high-pass frequency filtering (hard circular FFT mask, $r=40$), and YOLOv8l-Pose keypoints (34-dimensional normalised vector). *Spatial-statistical*: pixel shuffle, patch shuffle at four scales (2×2 – 16×16), and mean RGB. *Semantic*: Mask2Former segmentation (Swin-Large, ADE20K), YOLOv8x object detection (OpenImages V7, demographic categories remapped), BLIP Large captions (gender-neutral remapped), and SDv1.5 VAE reconstruction. Occlusion variants follow Meister et al.’s five-condition protocol: Full NoBg, MaskSegm, MaskSegm NoBg, MaskRect, and MaskRect NoBg, with person masks from Mask2Former COCO (96.5% detection rate, highest of five candidates).

3.4 Inference-Time Debiasing Pipeline

A 10,000-image debiased corpus was generated for the doctor profession at $s=7.0$, fully activating the model’s demographic prior as a demanding test of the debiasing interventions. ITI-GEN [22] inclusive tokens were trained on CCv2 across gender and skin tone (MST 1–10), producing 20 demographic combinations of 500 images each, applied via the `prompt_prepend` mechanism. ControlNet OpenPose [23] (scale 0.6, fixed canonical skeleton) enforced structural consistency and eliminated pose-correlated annotation confounds. Chamfer-based colour alignment [24] steered the predicted clean sample $\hat{x}_0$ at timesteps $t>600$ toward a skin-tone-neutral mean RGB reference. LCM-LoRA was excluded due to incoherent outputs under per-step Chamfer modification; SW-Guidance was excluded on throughput grounds.

4 Evaluation

4.1 Demographic Distribution Analysis

Both datasets exhibit a pronounced male skew across 200 professions, but the imbalance is substantially more severe in SDv1.5. Re-LAION-5B contains 41.5% female-labelled images (58.5% male), a 17 percentage point gap relative to parity; SDv1.5 widens this to 42 percentage points with only 29.02% female images. Against BLS 2024 and ILOSTAT baselines, Re-LAION-5B falls below real-world female proportions relative to US norms while remaining broadly consistent with global workforce proportions in several ISCO groups. SDv1.5 amplifies this skew across nearly every occupational group regardless of baseline, confirming a two-stage bias pattern in which the generative model exacerbates rather than inherits training corpus imbalances. Gender amplification is confirmed in 178 of 200 professions, with extreme cases including Logger (97.5% male), Nurse (97.5% female), and Receptionist (97.3% female).

The skin tone analysis reveals a near-complete collapse of dark-toned representation in generative outputs: the Dark-bin proportion falls from 0.81% in Re-LAION-5B to 0.14% in SDv1.5 (83% reduction), while the Light-bin proportion nearly doubles from 12.45% to 22.87%. Skin tone amplification toward lighter tones is confirmed in 175 of 200 professions. A real-world comparison was not conducted as the FairFace-derived race-to-skin-tone mapping collapsed all racial groups to the Mid bin.

4.2 Dataset Classification

A ConvNeXt-Tiny classifier distinguishes Re-LAION-5B, SDv1.5, and COCO with high accuracy across every transformation, from unmodified images at 99.42% down to mean RGB at 64.19%. A consistent discriminating hierarchy emerges: frequency and fine spatial statistics (high-pass 98.64%, patch shuffle 8×8 at 97.91%); structural geometry (depth 96.78%, Canny edge 93.62%); dense semantics (segmentation 91.36%, captions 90.99%, object detection 88.62%); and global colour statistics (pixel shuffle 80.67%, mean RGB 64.19%). No single transformation eliminates separability, confirming that dataset identity is redundantly encoded across colour, texture, geometry, semantics, and language simultaneously. VAE reconstruction reduces accuracy by only 0.27 percentage points (99.15% vs. 99.42%), ruling out low-level acquisition artefacts as the dominant signal source. Accuracies substantially exceed the 82.0% reference reported by Zeng et al. for purely web-scraped datasets,

consistent with SDv1.5’s generative pipeline imposing far greater internal homogeneity than a web-scraped corpus.

4.3 Demographic Artifact Probing

The occlusion experiments reveal a fundamental asymmetry. In Re-LAION-5B, all person-hidden variants collapse to chance for both gender and skin tone regardless of whether background is retained, establishing that background context carries no recoverable demographic signal. In SDv1.5, background context is a substantial independent carrier: with the person masked and background retained, gender AUC reaches 0.747 and skin tone AUC 0.728; silhouette alone (MaskSegm NoBg) reaches gender AUC 0.834. This directly replicates and substantially amplifies the contextual artifact identified by Meister et al.

Beyond occlusion, both datasets exhibit modest signal in non-appearance representations. Mean RGB achieves gender AUC 0.600/0.579 and skin tone AUC 0.563/0.543 (Re-LAION-5B / SDv1.5), consistent with the 0.580 reported by Meister et al. on COCO. Pose keypoints achieve gender AUC 0.697/0.680, confirming body configuration differences contribute independently of photometric properties. Object detection reveals gender-stereotyped scene compositions at AUC 0.692 on Re-LAION-5B (0.608 restricted), with the gap narrowing substantially in SDv1.5. Caption probing reveals a cross-dataset reversal absent from all visual probes: Re-LAION-5B achieves gender AUC 0.901 and skin tone AUC 0.749 versus SDv1.5 at 0.771 and 0.673, indicating that web-scraped images carry richer linguistically recoverable associations while SDv1.5 encodes stronger signal in scene-level visual statistics. Together, Re-LAION-5B suggests demographic signal is primarily localised within the person region while SDv1.5 distributes it pervasively across scene context, body geometry, object composition, and linguistic register.

4.4 Debiasing Evaluation

The combined ITI-GEN, ControlNet, and Chamfer colour alignment pipeline produces a substantive departure from the standard SDv1.5 demographic and spurious feature profile, achieving the intended 50% / 50% gender split and uniform spread across all ten MST levels. Person-level probing confirms successful conditioning: gender AUC reaches ≈ 1.000 on unmodified images (expected, as labels derive from the generation configuration).

Most non-person probes collapse to at or near chance, indicating substantial suppression of person-level demographic cues: mean RGB 0.507, object detection

0.449/0.457, depth at keypoints 0.454, pose keypoints 0.475, captions 0.530 (down 24.1 pp from the SDv1.5 value of 0.771). Skin tone non-person probes show the same pattern: mean RGB 0.482, object detection 0.439/0.461, depth 0.443, pose 0.483, captions 0.490, all at or near the 10-class chance level of 0.500.

However, background-retained occlusion (MaskRect) remains highly predictive (gender AUC 0.995; skin tone AUC 0.632), demonstrating that scene-level correlations persist despite debiasing.

The mean RGB skin tone value of 0.482 provides the clearest mechanistic confirmation that Chamfer alignment successfully disrupts photometric spurious correlations. The persistence of MaskRect performance indicates that scene composition remains partially entangled with demographic conditioning even under pose and appearance control, representing the primary open challenge for future work.

5 Conclusions and Future Work

This thesis applied and extended established spurious feature detection and demographic probing methodologies to Re-LAION-5B and Stable Diffusion v1.5, producing a systematic empirical characterisation of how demographic bias is encoded, distributed, and amplified across a web-scale training corpus and the generative model trained on it.

Addressing Objective 1, a profession-conditioned corpus of 200,000 images was retrieved from Re-LAION-5B and a corresponding 200,000-image generation corpus was produced using SDv1.5 with LCM-LoRA acceleration, both filtered through a two-stage structural validation pipeline. Demographic annotation was performed using the Realistic Gender Classifier and a VGG-16 MST10 Regression model evaluated against CCv2, MSTE, and FACET benchmarks, extending skin tone analysis to the full ten-point Monk Skin Tone scale beyond the binary and six-point scales used in prior work. Addressing Objective 2, the transformation-based framework of Zeng et al. was extended to a three-class Re-LAION-5B / SDv1.5 / COCO configuration across fifteen transformations, with every transformation exceeding the 33.3% chance level, confirming that dataset-specific spurious signatures persist under extreme information compression and are redundantly encoded across colour, texture, geometry, semantics, and language simultaneously. Addressing Objective 3, demographic probing following the occlusion framework of Meister et al. revealed a fundamental asymmetry between datasets: in Re-LAION-5B, demographic signal appears to be predominantly localised within person appearance, with background context contributing negligibly; however, the results do not permit a definitive conclusion that person-level features alone are sufficient. In contrast, SD v1.5 exhibits a fundamentally different structure: demographic information is distributed across scene context and background, recoverable from background context alone (gender AUC 0.747; skin tone AUC 0.728), from person silhouette geometry (gender AUC 0.834), and from natural language descriptions (gender AUC 0.771). Gender amplification was confirmed in 178 of 200 professions and skin tone amplification toward lighter tones in 175 of 200 professions. Addressing Objective 4, the combined ITI-GEN, ControlNet OpenPose, and Chamfer colour alignment pipeline successfully overrides SDv1.5’s strongly skewed learned prior, achieving a uniform demographic distribution by construction with most non-person probes collapsing to at or near chance for both gender and skin tone. Residual background-level signal persists (gender MaskRect AUC 0.995; skin tone MaskRect AUC 0.632), indicating that scene composition remains partially entangled with the demographic conditioning signal even under pose and appearance control. Addressing Objective 5, the combined evidence characterises a two-stage bias

propagation process in which Re-LAION-5B exhibits person-localised demographic signal that diverges from real-world labour force composition, and SDv1.5 then amplifies these imbalances substantially while transforming their distribution, projecting demographic priors beyond the person onto background, scene composition, object layout, and textual description. Inference-time debiasing can measurably reduce spurious correlations at the representation level without model retraining, but background-level entanglement represents the primary remaining challenge.

The most immediate future direction is targeted background conditioning to decorrelate scene composition from demographic attributes, whether through adversarial debiasing, extension of ITI-GEN to background properties, or weight editing via Unified Concept Editing [25], with a systematic comparison of conditioning-based versus weight editing approaches on the same probing protocol. Further extensions include evaluating the debiasing pipeline across a stratified sample of ISCO-08 major groups to assess generalisation beyond the doctor profession; validating debiased skin tone fidelity using automated MST estimation rather than generation-configuration labels; evaluating larger probe architectures or foundation model representations to determine whether the separability ceiling reflects dataset differences or probe capacity; and extending the analysis to a larger shard sample of Re-LAION-5B to assess the stability of identified spurious feature profiles across the full corpus.

References

- [1] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, “High-resolution image synthesis with latent diffusion models,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2022, pp. 10 684–10 695.
- [2] C. Schuhmann et al., *Laion-5b: An open large-scale dataset for training next generation image-text models*, 2022. arXiv: 2210.08402 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2210.08402>
- [3] A. Birhane, V. U. Prabhu, and E. Kahembwe, *Multimodal datasets: Misogyny, pornography, and malignant stereotypes*, 2021. arXiv: 2110.01963 [cs.CY]. [Online]. Available: <https://arxiv.org/abs/2110.01963>
- [4] A. S. Luccioni, C. Akiki, M. Mitchell, and Y. Jernite, *Stable bias: Analyzing societal representations in diffusion models*, 2023. arXiv: 2303.11408 [cs.CY]. [Online]. Available: <https://arxiv.org/abs/2303.11408>
- [5] F. Bianchi et al., “Easily accessible text-to-image generation amplifies demographic stereotypes at large scale,” in *Proceedings of the 2023 ACM Conference on Fairness, Accountability, and Transparency*, ser. FAccT ’23, Chicago, IL, USA: Association for Computing Machinery, 2023, pp. 1493–1504, ISBN: 9798400701924. DOI: 10.1145/3593013.3594095 [Online]. Available: <https://doi.org/10.1145/3593013.3594095>
- [6] N. Meister, D. Zhao, A. Wang, V. V. Ramaswamy, R. Fong, and O. Russakovsky, *Gender artifacts in visual datasets*, 2023. arXiv: 2206.09191 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2206.09191>
- [7] B. Zeng, Y. Yin, and Z. Liu, *Understanding bias in large-scale visual datasets*, 2024. arXiv: 2412.01876 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2412.01876>
- [8] F. Friedrich et al., *Fair diffusion: Instructing text-to-image generation models on fairness*, 2023. arXiv: 2302.10893 [cs.LG]. [Online]. Available: <https://arxiv.org/abs/2302.10893>
- [9] LAION, *Releasing re-laion-5b: Transparent iteration on laion-5b with additional safety fixes*, <https://laion.ai/blog/relaion-5b/>, 2024.
- [10] C. Schuhmann, *LAION-Aesthetics*, <https://laion.ai/blog/laion-aesthetics/>, Aug. 2022.
- [11] J. Ho, A. Jain, and P. Abbeel, *Denoising diffusion probabilistic models*, 2020. arXiv: 2006.11239 [cs.LG]. [Online]. Available: <https://arxiv.org/abs/2006.11239>

- [12] A. Radford et al., *Learning transferable visual models from natural language supervision*, 2021. arXiv: 2103.00020 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2103.00020>
- [13] J. Ho and T. Salimans, *Classifier-free diffusion guidance*, 2022. arXiv: 2207.12598 [cs.LG]. [Online]. Available: <https://arxiv.org/abs/2207.12598>
- [14] D. Podell et al., *Sdxl: Improving latent diffusion models for high-resolution image synthesis*, 2023. arXiv: 2307.01952 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2307.01952>
- [15] International Labour Organization, *International standard classification of occupations (ISCO-08)*, Accessed: 2025, 2012. [Online]. Available: <https://ilostat.ilo.org/methods/concepts-and-definitions/classification-occupation/>
- [16] R. Naik and B. Nushi, *Social biases through the text-to-image generation lens*, 2023. arXiv: 2304.06034 [cs.CY]. [Online]. Available: <https://arxiv.org/abs/2304.06034>
- [17] V. Matias and J. Neto, “Enhancing fairness in machine learning: Skin tone classification using the monk skin tone scale,” Sep. 2024, pp. 76–81. DOI: 10.5753/sibgrapi.est.2024.31648
- [18] C. Schumann, G. O. Olanubi, A. Wright, E. M. Jr., C. Heldreth, and S. Ricco, *Consensus and subjectivity of skin tone annotation for ml fairness*, 2024. arXiv: 2305.09073 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2305.09073>
- [19] B. Porgali, V. Albiero, J. Ryda, C. C. Ferrer, and C. Hazirbas, *The casual conversations v2 dataset*, 2023. arXiv: 2303.04838 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2303.04838>
- [20] L. Gustafson et al., *Facet: Fairness in computer vision evaluation benchmark*, 2023. arXiv: 2309.00035 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2309.00035>
- [21] J. Buolamwini and T. Gebru, “Gender shades: Intersectional accuracy disparities in commercial gender classification,” in *Proceedings of the 1st Conference on Fairness, Accountability and Transparency*, S. A. Friedler and C. Wilson, Eds., ser. Proceedings of Machine Learning Research, vol. 81, PMLR, 23–24 Feb 2018, pp. 77–91. [Online]. Available: <https://proceedings.mlr.press/v81/buolamwini18a.html>
- [22] C. Zhang et al., *Iti-gen: Inclusive text-to-image generation*, 2023. arXiv: 2309.05569 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2309.05569>

- [23] L. Zhang, A. Rao, and M. Agrawala, *Adding conditional control to text-to-image diffusion models*, 2023. arXiv: 2302.05543 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2302.05543>
- [24] K. C. Shum, B.-S. Hua, D. T. Nguyen, and S.-K. Yeung, *Color alignment in diffusion*, 2025. arXiv: 2503.06746 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2503.06746>
- [25] R. Gandikota, H. Orgad, Y. Belinkov, J. Materzyńska, and D. Bau, *Unified concept editing in diffusion models*, 2024. arXiv: 2308.14761 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2308.14761>